Project 1 Data Card

Exploring Tea Production and Nutritional Context through Data Analysis

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Card ID

Primary Dataset	FAO Tea Production Dataset
Description	Contains yearly country-level tea production quantities reported to the Food and Agriculture Organization (FAO), which shows global tea production trends over decades
Secondary Dataset	Daily Food & Nutrition Dataset
Description	Dataset contains daily dietary intake records, and this includes nutritional values for foods and drinks such as tea

Add more rows to the above table for each additional supplemental dataset.

Datasets at a glance

FAO Tea Production Dataset

Dataset contains yearly tea production quantities reported by countries to the Food and Agriculture Organization (FAO) of the United Nations.

Fields (if applicable)

Field name	Data Type	Description
Country	Categorical (String)	Short description, value range, and example values
Year	Numerical (Integer)	Year reported
Production_Quantity	Numerical (Decimal)	Tea production quantity (typically measured in metric tonnes)

Daily Food & Nutritional Dataset

The supplemental dataset contains generated daily dietary records representing food and drink consumption, including nutritional values for tea and other drinks as well. The dataset does not correspond to real individuals and is mainly used for analysis.

Field name	Data Type	Description
Food_Item	Categorical (String)	Short description, value range, and example values
Category	Categorical (String)	Food category
Calories	Numerical (Float)	Energy content per serving
Protein	Numerical (Float)	Protein content per serving
Sugar	Numerical (Float)	Sugar content per serving
Meal_Type	Categorical (String)	Meal context

Additional notes

- This dataset does not differentiate between tea subtypes

Usage information

Use(s)

- Analyzing global trends in tea production
- Comparing tea production across regions
- Giving context for tea as a drink

Licensing

- FAO Tea Production Dataset: FAO Open Data License (free to use with attribution)
- Daily Food & Nutrition Dataset: Kaggle Dataset License (non-commercial, attribution required)

Privacy considerations

- The FAO dataset contains no personal or biased data, pretty much only statistics and numbers
- The nutrition dataset also contains no personal data or PII

Format

The primary dataset is in a [**CSV file** | JSON file | …].
The supplemental dataset is in a [**CSV file** | JSON file | …].].

Source and collection process

Origin

- The FAO Tea Production dataset is from the FAO database
- The nutrition dataset was generated using publicly available food references

Original datasets author/publisher

- FAO Tea Production: Food and Agriculture Organization of the United Nations
- Nutrition Dataset: Kaggle Contributor

Original datasets point of contact

- FAO Statistics Division
- Kaggle dataset discussion page

Original metadata

- FAO metadata available via FAOSTAT documentation
- Dataset description available on Kaggle dataset page

Collection process

FAO: National agricultural reporting -> FAO standardization
Nutrition dataset: Simulation based on typical nutrition ranges

Collection or access date

FAO dataset downloaded: 1/29
Nutrition dataset downloaded: 1/29

Refresh date/schedule

FAO dataset: Updated yearly
Nutrition dataset: One-time synthetic release (no refresh schedule)

Exploration

Questions driving the exploration study

1) What countries have been the leading producers of tea over the period of time?
2) How has tea production changed over time across different regions?
3) How does tea compare nutritionally to other low-calorie drinks?
4) What is the nutritional profile of tea (overall)?
5) To what extent does tea production in 1990 correlate with tea production in 2021?

Descriptive statistics

Tea Production Dataset:

Country: **Count:** This shows multiple tea-producing countries, representing a broad global sample of tea production

Year: Range: The summary stats were calculated for the 2021 production column. The mean production value is higher than the median, which indicates a right-skewed distribution

Production Quantity: Max Value: This identifies the single largest production record in the dataset, in case the single largest tea-producing country

Daily Food & Nutrition Dataset:

Calories: Mean/SD: The mean and SD summarize the overall distribution of calorie values across food and drinks

Sugar: Median Value: The median sugar value represents the typical sugar content within the dataset

Protein: Max Value: The maximum protein value highlights the upper range of protein content among food items

Distribution plots

Food Category Distribution: A count plot shows the relative frequency of different food categories in the dataset

Tea Production Distribution: A bar plot illustrates the distribution of tea production across top producing countries

Protein Distribution: A histogram displays the spread of protein content values

Fat Distribution: A histogram displays the spread of fat values

Visual analysis

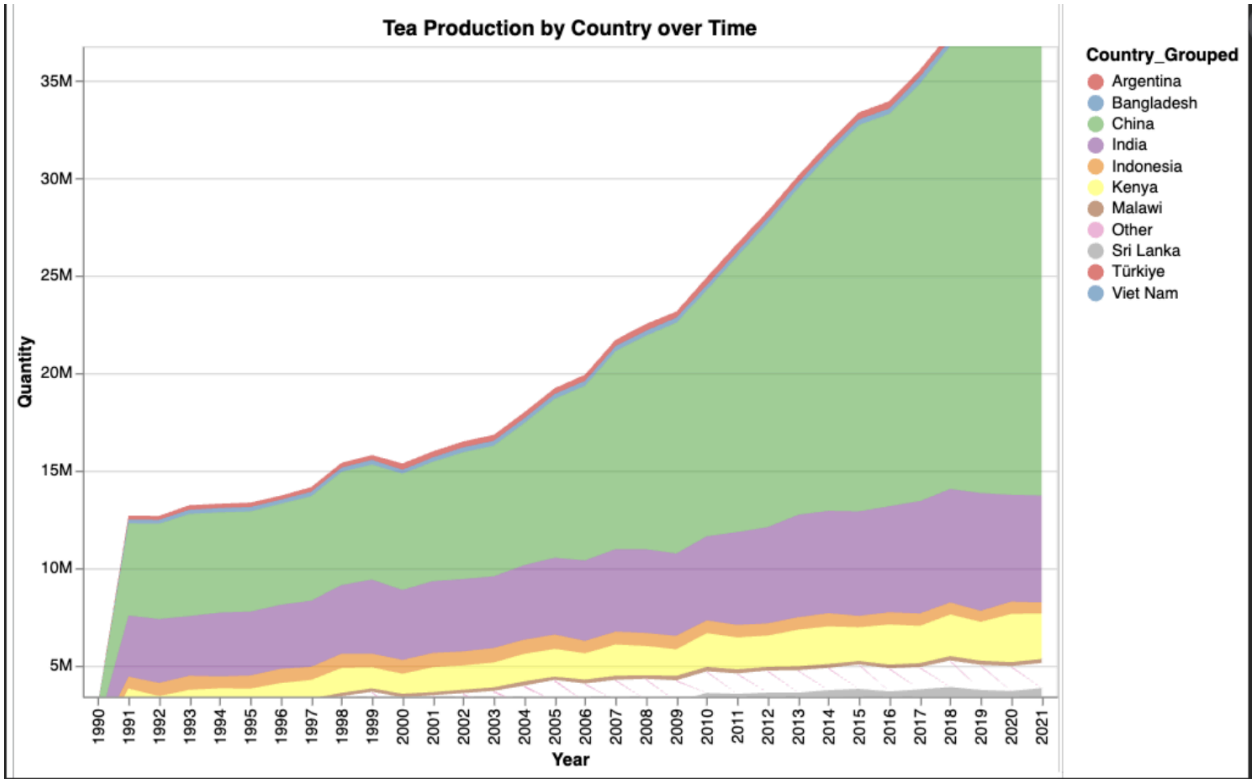

Chart 1: Tea Production by Country over Time (Stacked Area Chart)

- This visualization shows how the global production has tripled since 1991, and that China (green section) accounts for the majority of that tea production growth.
- Other major players such as India (purple) and Kenya (yellow) show pretty consistent growth as well, but their proportions remain stable compared to China's.
- The total quantity also shows upward growth from like 12M in 1991 to over 35M by 2021, showing a massive increase in global tea demand as well.

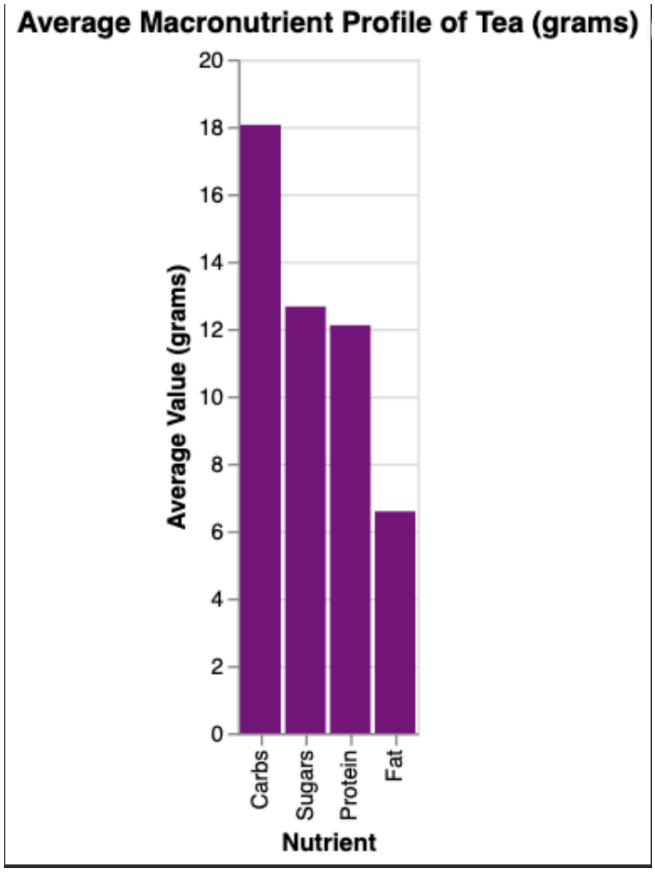

Chart 2: Average Macronutrient Profile of Tea (Bar Chart)

- This profile revelas how carbs are the primary macronutrient in tea, averaging around 18 grams per unit served
- We found it surprising how sugars and protein both sit at very similar levels (around 12 and 13 grams). We thought that sugar would be the leading factor!
- Fat is the least prevalent macronutrient at around 7 grams, which goes pretty well with tea's reputation of being a low-fat beverage

Analytical & statistical methods

1) Correlation Analysis (scatterplot): A analysis was done on 1990 and 2021 production values. This revealed a decently strong positive correlation, which indiciated that a country's past output is a reliable predictor of its modern day scale
2) Time-Series Analysis (stacked area chart): Production data was aggregated by both year and region to track the cumulative growth. We saw that the global output tripled over 30 years, and the growth was also non uniform which was mainly China's exponential expansion with tea production
    a) We also used time series analysis to find the leading countries of tea production and how much they produced overall (regular multiple line graph)

3) Descriptive Statistics (bar chart): Means (averages) were calculated for various macronutrients (for tea, specifically) to make up a nutritional profile. This showed us a hierarchy of macronutrients with Carbohydrates (~18 grams) being the most significant component compared to fats. We found that proteins and sugars were around the same (this was surprising to us)
4) Comparative Analysis: We normalized multiple continuous variables (such as calories, fat, protein) to compare tea against other beverages. This highlighted tea's specific strengths and weaknesses for us to compare against other drinks.

Cleaning and processing

Missing values

Identification: Used .isnull().sum() to identify missing entries in both Nutrition and Tea datasets
Nutrition Data Method: Applied group based mean imputation. Missing values for calories and macronutrients were filled using the mean (average) of their specific food category to maintain category specific nutritional profiles.
Tea Data Method: Used linear interpolation within country-specific groups. This method was chosen because tea production is time series data (and after doing some research, we found out that interpolation estimates missing years by drawing a logical trend line between known data points)
Verification: Performed a final null count check to ensure 100% data completeness before moving to the join stage.

Aggregation

Tea Production: Aggregated data using the multiindex level 'country' to calculate the mean production and smoothed production across the 2020-2021 timeframe.
Nutrition Data: Grouped the dataset by Category_Enum to provide a summary of the average nutrient density per good group.
Smoothing: Created a 3 year rolling mean to aggregate and condense local fluctuations and highlight the overall production trend for each country.

Joins of input sources

Join Strategy: Executed a left join using a "cross-join" logic via a temporary merge_key
Justification: Because the primary and supplementary datasets did not share a common ID, we isolated a separate representative nutritional profile for "Tea" and then joined it to all production records.
Result: This then merged the time-series production data with nutritional benchmarks, creating a consolidated Dataframe in parquet format for analysis